Airport and Flight Management Project Documentation

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1.What is it and what functionalities does it have?

"Airport and Flight Management" is a basic model of platform of airport management. Through this webpage, some simple, but useful functionalities can be done.

Before any further information, it is important to make clear the way in which the whole webpage it is divided: the main page is called "Airport and Flight Management"; from it, the user has access to other secondary pages which are numbered from 1-4 below. Each of these offer access to more others (labeled with letters) which have suggestive names (in conformity with their functionality):

1. FLIGHT INFORMATION – there can be found all administrative information regarding the flights (from the past and future), such as: their id, the airlines that operate them and their codes, the scheduled dates, take off time and landing time, departure cities and arrival cities. This information is classified in conformity with the current date, therefore the data of those flights which already happened are on the "Previous Flights Info" page, whereas the upcoming ones are on the "Upcoming Flights Info" page. Moreover, from "Flight Information", the user can access "Flight Personnel Info" page, where all the names of each cabin crew for all flights can be found, from both, past and future.

a. PREVIOUS FLIGHTS INFO

b. UPCOMING FLIGHTS INFO

c. FLIGHT PERSONNEL INFO

2. SHIFTS –here, people that work for the airport can check the past/future-programmed shifts on "View Personnel Shift" and introduce information about their past shifts or program one in the future on "Introduce Shift Details"

a. VIEW PERSONNEL SHIFTS

b. INTRODUCE SHIFT DETAILS

3. PASSENGER INFORMATION – just like on "Flight Information" page, here are other 2 pages classified concerning the time aspect: "Previous Passengers" and "Upcoming Passengers". Moreover, it aims offering information about passengers (the reservation number, name, CNP, the day of the flight they have been on or on which they will be, the take-off time, the landing time, the departure and arrival city and the information about the seat row and seat letter), the ones who travelled in the past and the ones who already made reservations for flights scheduled in the future. Also, by the reservation number of a passenger, they can be found together with the flight information on "Search Passenger by Reservation Number" (same information as in previous and upcoming passengers but another order).

a. PREVIOUS PASSENGERS

b. UPCOMING PASSENGERS

c. SEARCH PASSENGER BY RESERVATION NUMBER

4. BOOK A FLIGHT – First of all, this provides a view of the flights which still have available spots - the number of them, and the number of the occupied ones, on “Available Flights” page. In addition, it gives a list of the non-available flights (the status “FULL” being mentioned) on “Non-Available Flights”, and gives the user the opportunity to make a reservation. The pages with the table contain the same data as in “Flight Information”, but with the additional fields mentioned above. Also, for making a reservation, the user has to introduce the name of the person whose ticket belongs to, their CNP, home-country, and the departure and arrival cities (one of them MUST be Cluj-Napoca) on “Make a Reservation” page.

a. MAKE A RESERVATION

b. AVAILABLE FLIGHTS

c. NON-AVAILABLE FLIGHTS

2.How does it work?

This is the main page, main menu, from where the user can choose between 4 main classes of functionalities.



Each button will redirect the user to one of the further “secondary menus”.

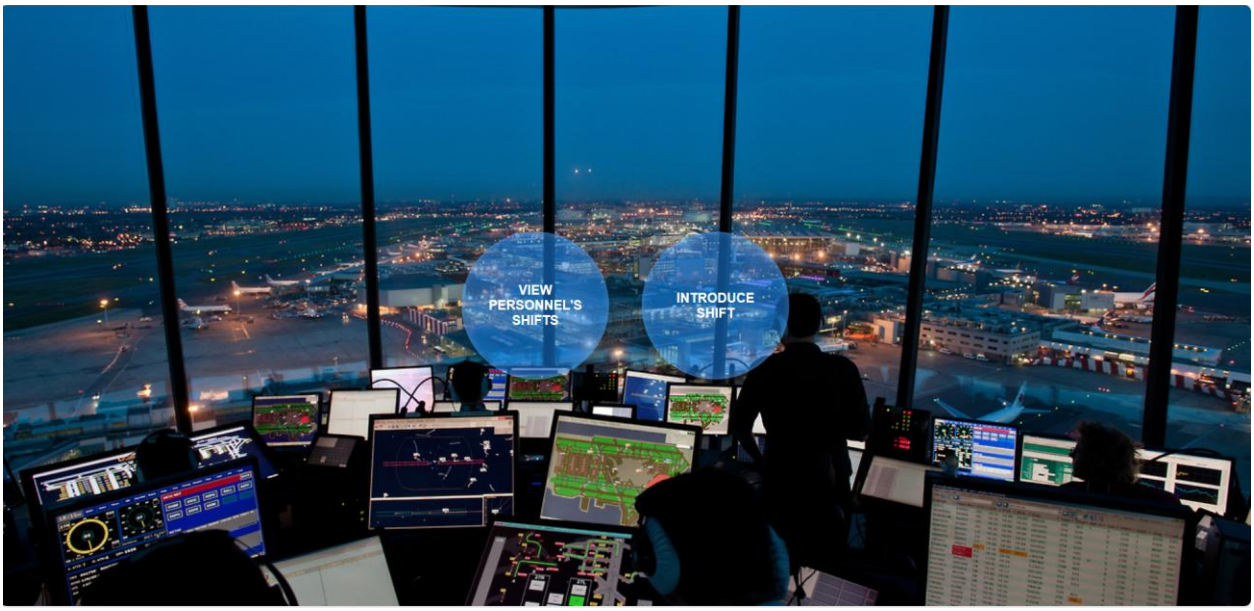
1. Flight Information



2. Book A Flight



3. Shift Details



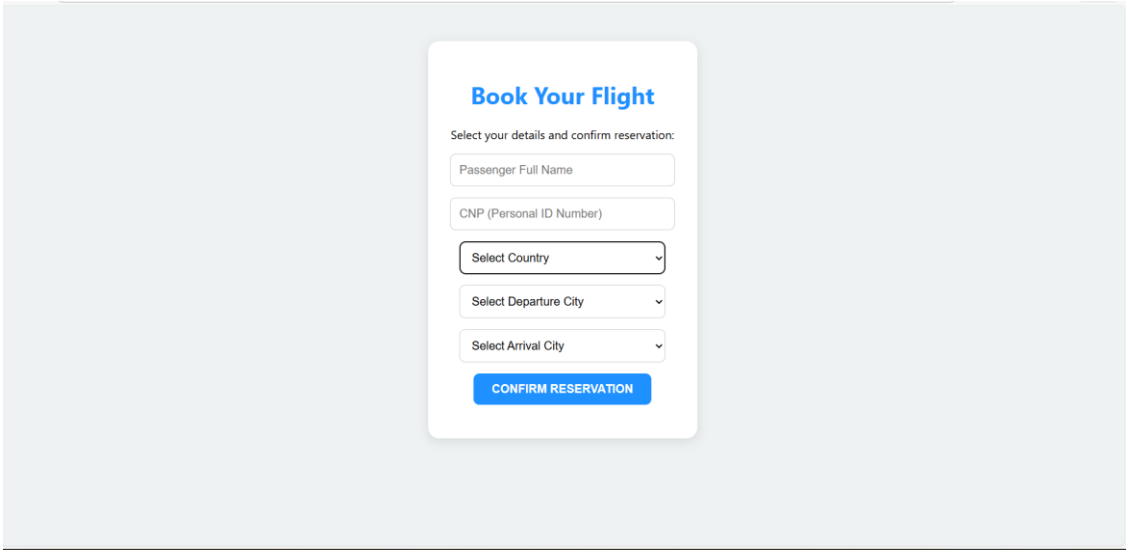
4. Passenger Information

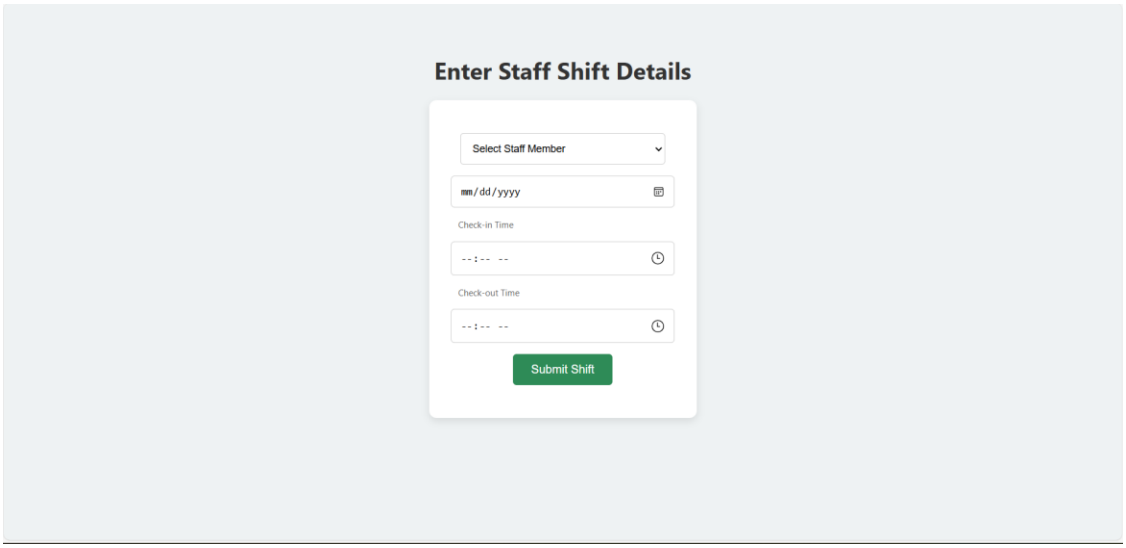


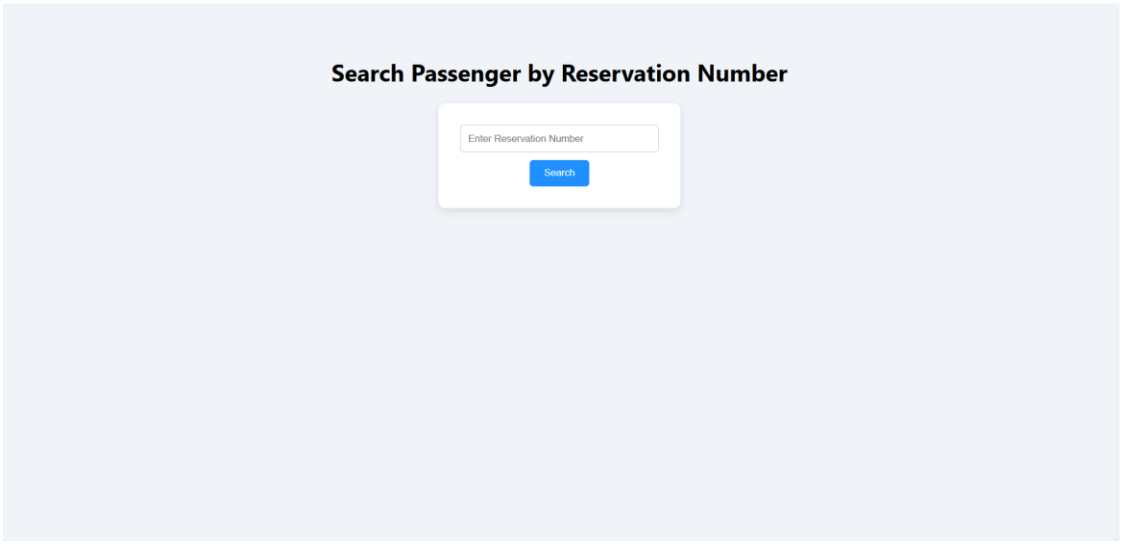
From each “secondary menu” the user can access, through buttons, the wanted tables for viewing or proceed insertions/searches. These are some examples of expected forward pages:

PREVIOUS FLIGHTS

Airline Code	Flight ID	Departure City	Arrival City	Date	Take-Off	Landing	Airline Name
A	1001	Cluj-Napoca	Bucharest	2025-12-01	09:00:00	10:00:00	AnimaWings
A	1002	Bucharest	Cluj-Napoca	2025-12-01	10:30:00	11:30:00	AnimaWings
LH	1003	Cluj-Napoca	Nice	2025-12-03	12:30:00	15:00:00	Lufthansa
LH	1004	Nice	Cluj-Napoca	2025-12-03	16:45:00	19:30:00	Lufthansa
H	1005	Cluj-Napoca	Paris	2025-12-01	18:30:00	19:30:00	HiSky
H	1006	Paris	Cluj-Napoca	2025-12-01	20:00:00	22:30:00	HiSky
LH	1007	Cluj-Napoca	Munich	2025-12-02	16:30:00	18:30:00	Lufthansa
LH	1008	Munich	Cluj-Napoca	2025-12-02	18:45:00	20:45:00	Lufthansa
FR	1009	Cluj-Napoca	Milano	2025-12-03	00:40:00	02:30:00	Ryanair
FR	1010	Milano	Cluj-Napoca	2025-12-03	03:00:00	04:45:00	Ryanair
ROT	1011	Cluj-Napoca	Rome	2025-12-05	05:30:00	07:30:00	Tarom
ROT	1012	Rome	Cluj-Napoca	2025-12-05	08:00:00	09:45:00	Tarom


Book Your Flight
Select your details and confirm reservation:
Passenger Full Name
CNP (Personal ID Number)
Select Country
Select Departure City
Select Arrival City
CONFIRM RESERVATION


Enter Staff Shift Details
Select Staff Member
mm/dd/yyyy
Check-in Time
--:-- --
Check-out Time
--:-- --
Submit Shift


Search Passenger by Reservation Number
Enter Reservation Number
Search

3.What is it composed of?

First of all, my project is composed of a database, written in PostgreSQL, in which I have the tables that hold the information about the passengers, employees, flights etc. Moreover, it includes a Java project, through which it was made possible the connection between the front end and the back end of the website (using JSON as well), the interface being composed of multiple HTML files, written in HTML, CSS and JavaScript.

Second of all, this project includes a demo, where I tried to mimics the functionalities of the website in Java, the original Scheme of the project and a video of how it works and how it should look like https://youtu.be/cIYVLxrfOvs .

Furthermore, the project was meant to be used by the operators of the "Avram Iancu" International Airport from Cluj-Napoca, which have the authority and access to this specific kind of data. The website was not made for any kind of users. In addition, the way of organizing the information about the past and future flights and passengers, airport and airlines' personnel, was created for a better control and monitorization, in case of emergencies, police needs, crime cases or simply for a more systematic perspective of the airport.

4.Requirements

Below, I had listed all the tasks I had to respect and examples of passages in my project where I solved them:

1. Project Concept

- Hospital Management System

- Composition: AirportManager has lists of lists of cabin crew members stored in a list of type Airline Staff

-I will give examples based on Person class and its children

Attributes (fields): name, function_id, airline_id, cnp

Methods: getDescription

Relationships between classes (inheritance-mentioned below, aggregation-AirportManager manages collections of flights and people, composition-mentioned below)

2. Technical Requirements

-Encapsulation: all classes extended from Person have private fields and public methods: Staff, Passenger etc.

-Inheritance

Person-parent class, Staff and Passenger inherit the name from Person, Airline_Staff and Airport_Staff inherit the function_id from Staff and the name from Person

-Polymorphism

in Person there is the function getDescription which is Overridden in Airline_Staff, Airport_Staff, Passenger

-Constructor chaining

constructors use super for initializing the name of the subclasses of Person

-Method overriding

Flight-FlightDetails method or getDescription again

-Collection usage (arrays, ArrayList, HashMap, etc.)

ArrayList for storing flights in AirportManager, or HashMap for passenger look- up through reservation number

-Persistent data (via a database connection or file serialization)

DATABASE CONNECTION

-Basic frontend page (HTML or framework-based) for user interaction

make_a_reservation.html, insert_shift.html

-Menu-based or GUI interaction for adding, updating, and viewing data

web-based menu-webpage.html